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Meslioui

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(54) **TURBO-COMPOUNDED ENGINE WITH EXHAUST DUCT ACOUSTIC ARRANGEMENT**

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(51) **Int. Cl.**

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F01N 1/24	(2006.01)
F02K 1/82	(2006.01)
B64D 27/02	(2006.01)
F02C 7/24	(2006.01)

(52) **U.S. Cl.**

CPC ***B64D 33/06*** (2013.01); ***B64D 27/04*** (2013.01); ***F01N 1/24*** (2013.01); ***F02K 1/82*** (2013.01); ***F02K 1/827*** (2013.01); *B64D 27/026* (2024.01); *B64D 33/04* (2013.01); *F02C 7/24* (2013.01); *F05D 2260/963* (2013.01)

(58) **Field of Classification Search**

CPC B64D 33/06; B64D 27/04; B64D 27/06; F01N 1/24; F02K 1/82; F02K 1/827; F01D 25/30; F02C 7/24

See application file for complete search history.

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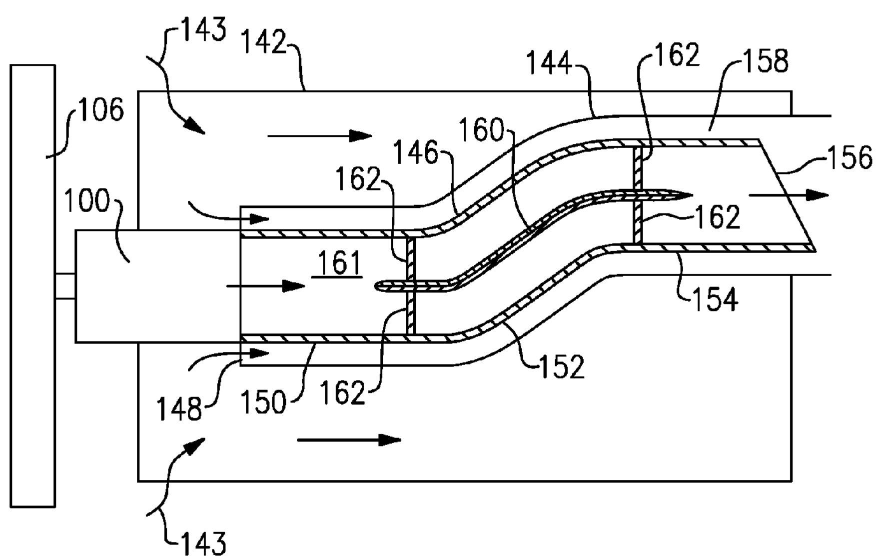
(57) **ABSTRACT**

A turbo-compounded engine includes a piston engine connected to drive a propulsor. An outlet of the piston engine is operable to connect products of combustion from the piston engine to pass over a turbine. The turbine is connected to drive a turbine shaft also connected to drive the propulsor. An outlet of the turbine is connected into an exhaust duct configured to exhaust the products of combustion. The exhaust duct is provided with an exhaust duct outer wall defining an exhaust chamber. A further cooling air outer wall is positioned outwardly of the exhaust duct. Flow dividers are received within an exhaust chamber inward of the exhaust duct outer wall. The exhaust duct outer wall has an inner surface and the flow dividers have an outer surface. Acoustic treatment is provided on both the inner surface of the exhaust duct outer wall and the outer surface of the flow dividers.

9 Claims, 5 Drawing Sheets



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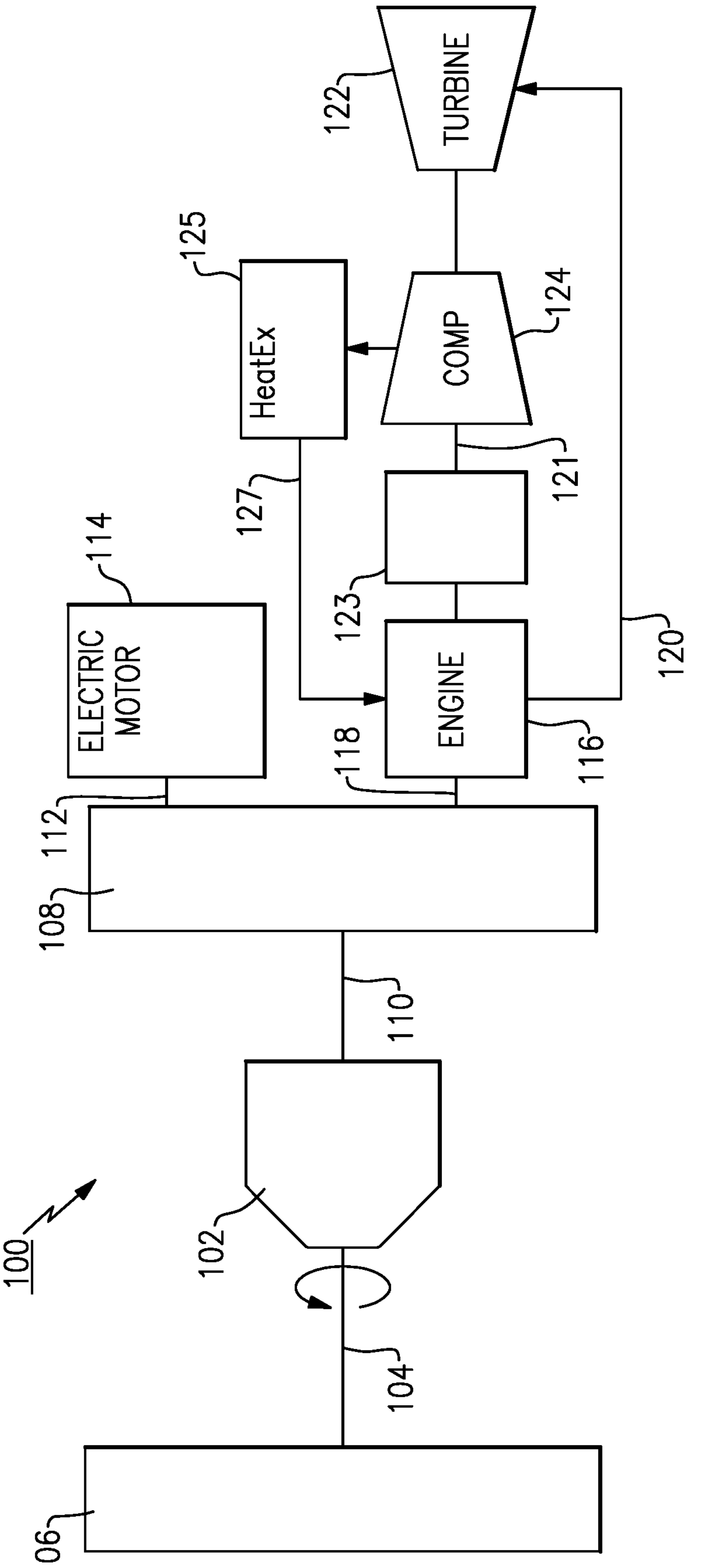
TURBINE
122
125
HeatEx
COMP
124
121
127
123
114
ELECTRIC MOTOR
ENGINE
120
116
118
112
108
110
102
100
104
106

FIG.1

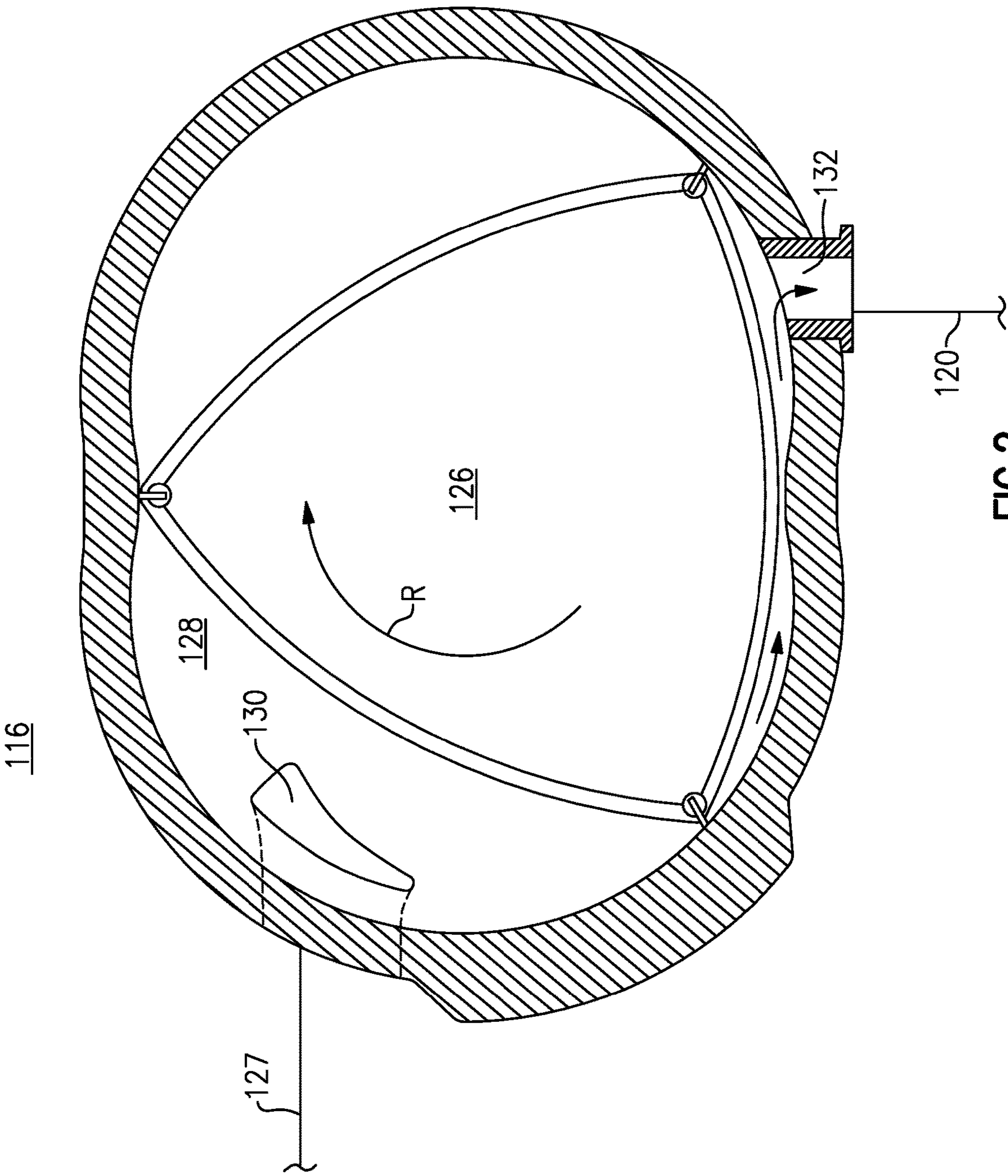

FIG.2

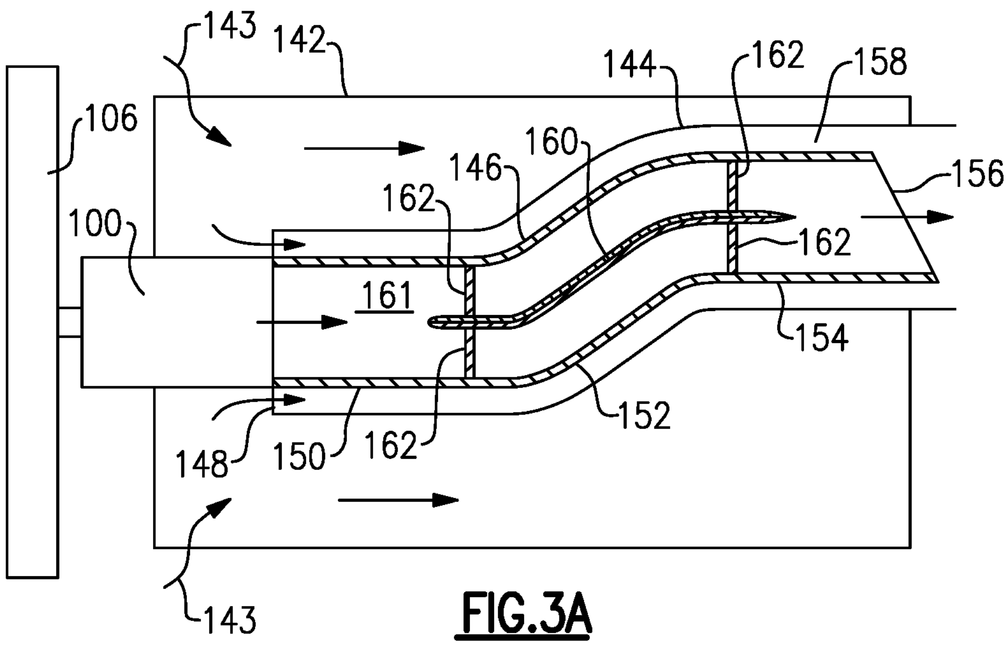
143
142
144
162
158
106
160
146
156
162
162
100
161
154
162
152
148
150
143

FIG.3A

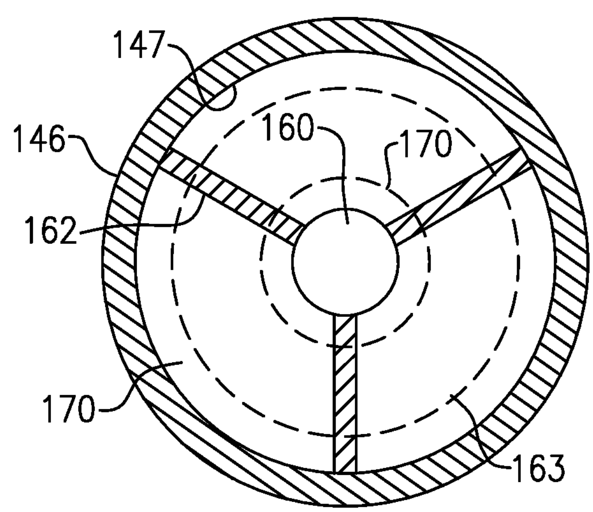
147
146
160
170
162
170
163

FIG.3B

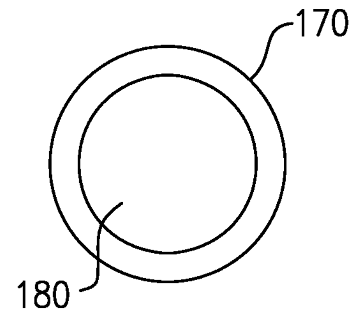

FIG.4A

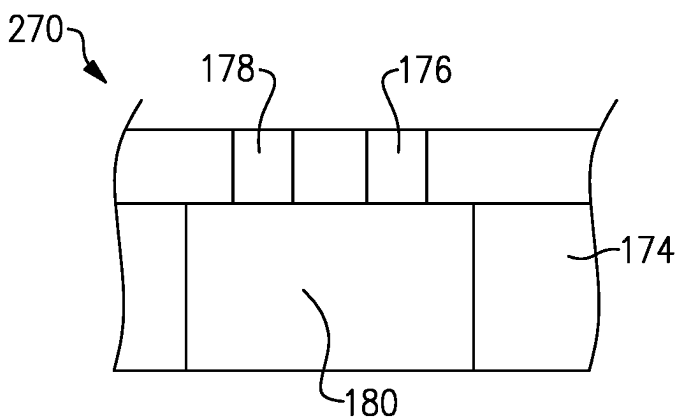

FIG.4B

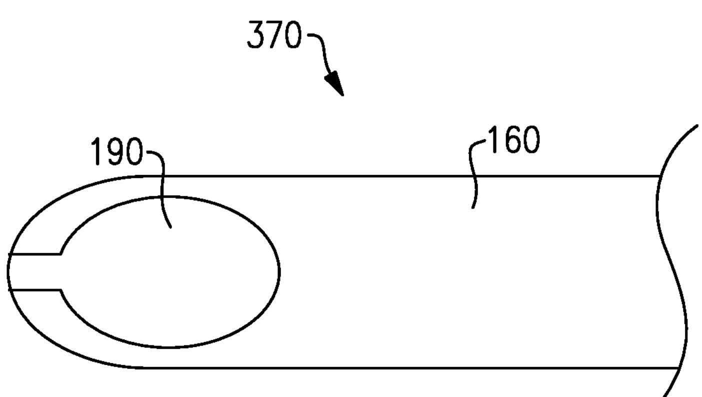

FIG.4C

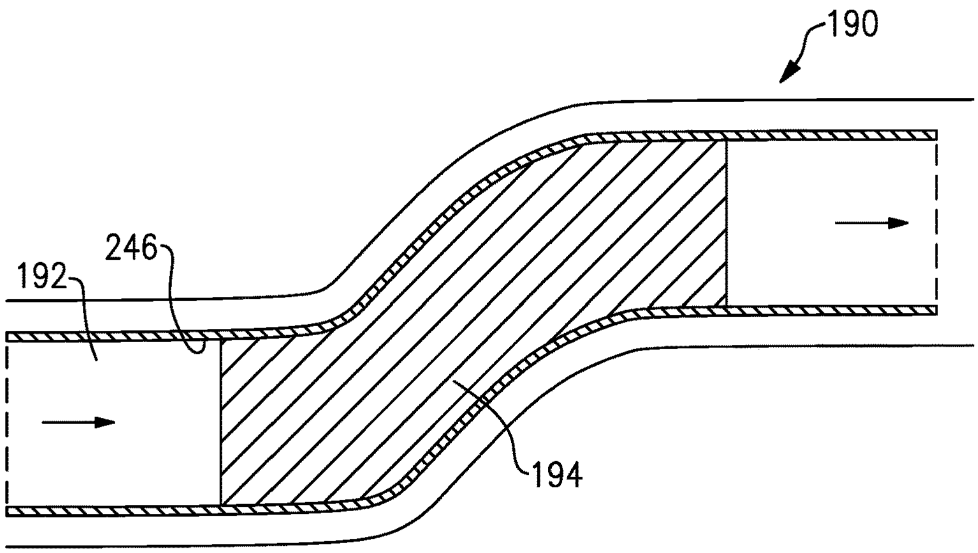

FIG.5A

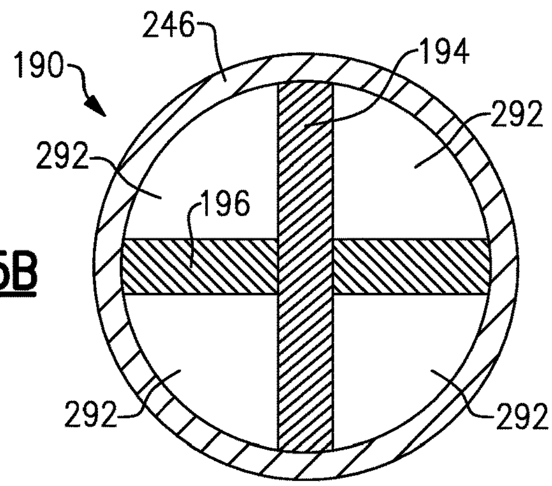

FIG.5B

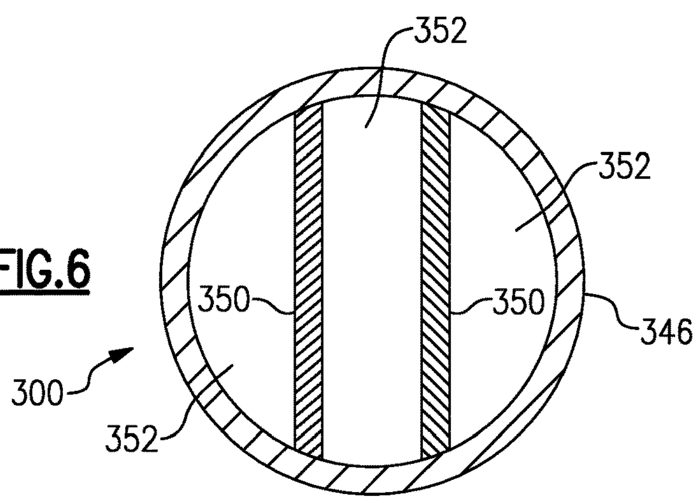

FIG.6

TURBO-COMPOUNDED ENGINE WITH EXHAUST DUCT ACOUSTIC ARRANGEMENT

BACKGROUND

This application relates to providing acoustic arrangements to reduce exhaust noise in a turbo-compounded engine.

Engines for aircraft applications typically include a propulsor. In some engines a fan is used and in others a propeller.

One type of engine that is showing some promise is a turbo-compounded engine. In a turbo-compounded engine there is typically a piston engine driving the propulsor. Exhaust products downstream of the piston engine pass across a turbine. The rotation of the turbine drives a shaft that also provides propulsion to the propulsor.

Such engines are provided with relatively long exhaust ducts to handle the noise that will be created. Further, the exhaust duct shape needs to fit particular installation challenges for a turboprop.

Another engine type is an auxiliary power unit ("APU"). These engines do not have a propulsor, but may use turbo-compounded engines. Such APU engines also have noise challenges.

In one type of turbo-compounded engine an electric motor also supplements the piston engine. The exhaust noise challenges in such engines are particularly acute.

SUMMARY

A turbo-compounded engine includes a piston engine connected to drive a propulsor. An outlet of the piston engine is operable to connect products of combustion from the piston engine to pass over a turbine. The turbine is connected to drive a turbine shaft also connected to drive the propulsor. An outlet of the turbine is connected into an exhaust duct configured to exhaust the products of combustion. The exhaust duct is provided with an exhaust duct outer wall defining an exhaust chamber. A further cooling air outer wall is positioned outwardly of the exhaust duct. Flow dividers are received within an exhaust chamber inward of the exhaust duct outer wall. The exhaust duct outer wall has an inner surface and the flow dividers have an outer surface. Acoustic treatment is provided on both the inner surface of the exhaust duct outer wall and the outer surface of the flow dividers.

These and other features will be best understood from the following drawings and specification, the following is a brief description.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** schematically shows a turbo-compounded engine.

FIG. **2** shows a piston engine incorporated into the turbo-compounded engine of FIG. **1**.

FIG. **3**A shows an exhaust duct arrangement for a turbo-compounded engine.

FIG. **3**B is a cross-section of the FIG. **3**A embodiment.

FIG. **4**A shows a first acoustic treatment.

FIG. **4**B shows a second example acoustic treatment.

FIG. **4**C shows yet another alternative acoustic treatment.

FIG. **5**A shows a distinct embodiment exhaust duct.

FIG. **5**B is a cross-section through FIG. **5**A.

FIG. **6** shows yet another exhaust duct embodiment.

DETAILED DESCRIPTION

A turbo-compounded engine **100** is illustrated in FIG. **1**. Engine **100** is for use on an aircraft. A gear reduction **102** is shown schematically driving a shaft **104** to in turn drive a propulsor **106**. Propulsor **106** rotates at a slower speed than shaft **100** due to gear reduction **102**. In embodiments the propulsor **106** could be a propeller. However, it is also within the scope of this disclosure that the propulsor could be a fan. A compound gearbox **108** takes in several sources of drive to in turn drive a shaft **110** that drives the gear reduction **102**.

An electric motor **114** optionally drives a shaft **112** to provide rotational force into the compound gearbox **108**.

A piston engine **116** drives a shaft **118** to also provide a source of rotation into the gearbox **108**. It is known that the electric motor can provide additional power in combination with the piston engine **116** under certain operational conditions such as high power conditions. As an example, at takeoff both the electric motor **114** and the piston engine **116** may be utilized. At lower power conditions, such as cruise, perhaps only the electric motor **114** is used.

A compound gearbox **123** receives rotation from a shaft **121**. An exhaust **120** from the piston engine **116** passes across a turbine **122**. The turbine **122** drives the shaft **121** through the gearbox **123** to provide additional rotational force to shaft **118**. The turbine **122** is also shown driving a compressor **124**. The compressed air from compressor **124** passes through a heat exchanger **125** and then to the piston engine **116** through line **127**.

FIG. **2** shows an example of the piston engine **116**. A rotary piston engine, such as a Wankel engine is illustrated. A piston **126** rotates within a chamber **128**. Compressed air at **130** is injected into the chamber **128** from line **127**. It is mixed with fuel and ignited and drives the piston **126**. The products of combustion pass through outlet **132** and into line **120**.

The operation of the engines **100** and **116** may be as known.

It is also known that a turbo-compounded engine could have a piston engine with a plurality of engines such as engine **116** as shown in FIG. **2**. Further, it is known that the piston engine could be a reciprocating piston.

As mentioned above, there are challenges with exhaust noise in such engines, because of the pulsation nature of the rotary core.

FIG. **3**A shows an exhaust for an engine such as engine **100**. A nacelle **142** surrounds a core housing **144**. The propulsor **106** provides propulsion for an associated aircraft. Propulsor **106** also delivers air into the nacelle **142** as shown at **143**. Some of the air passes outwardly of an inner housing **144** to cool the inner housing **144**.

While a propulsor is shown, the teachings of this disclosure extends to turbo-compounded engines in non-propulsor applications, such as APUs.

Another portion of the air is pumped into an inlet **148** defined between the housing **144** and an outer exhaust housing **146**. That air passes downstream and through an ejector opening **158**.

The products of combustion leaving the engine **100** enter a chamber **161** inward of housing **146** and provide a jet pump at an exit **156** of the housing **146**.

The housing **146** may be thought of as an S-duct arrangement. There is a radially inner first generally axially extending portion **150** leading into a radially outwardly extending portion **152**, which then merges into a second generally axially extending portion **154**.

The first generally axially extending portion **150** extends axially away from the engine **100** with a component in an axial direction defined by the rotational axis X of the engine. The axial component is greater than a radial component of its direction.

The first generally axially extending portion **150** merges into radially extending portion **152**. The radially extending portion **152** merges into the second generally axially extending portion **154**.

Portion **154** extends along a direction with a component in an axial direction that is greater than a radial component.

The radially extending portion **152** extends with an axial component but also having a radial component that is greater than the radial component in both the first and second axially extending portion **150/154**.

In an engine such as engine **100**, the S-duct must be made relatively long to provide sufficient area for exhaust noise mitigation and reduce exhaust noise. This is particularly true when an electric motor **114** is utilized in the engine **100**.

Applicant has recognized that breaking the exhaust gas flow in chamber **161** in subportions will help reduce the noise leaving the exhaust duct exit **156**.

Thus, a flow divider is provided by a center bullet **160** and struts **162**.

As shown in FIG. **3**B, there are three chambers **163** defined between circumferentially spaced ones of the struts **162**. The struts **162** may have an aero shape to minimize losses. The struts **162** and center bullet **160** thus provide flow dividers. Acoustic treatments **170** are shown on an inner surface **147** of outer wall **146** and an outer surface of the center bullet **160**. The thickness of treatments **170** is exaggerated for illustration purposes only. In practice they will likely be thinner.

FIG. **4**A shows a first embodiment of the acoustic treatment on the bullet **160**. The acoustic treatment **170** may be a coating of foam or other porous materials such as fiberglass or mesh. Mesh treatments include porous materials bagged in a mesh cloth and finished with a perforated sheet to form the skin facing the flow. This avoids contamination/deterioration with hot gases in the exhaust. Similar treatments can be provided on inner surface **147**.

Alternatively, FIG. **4**B shows an acoustic treatment **270** which includes a plurality of chambers **180** in a flow divider body **174**, and having a face plate **176** with small perforations **178** over the chamber **180**.

An acoustic treatment **370** as shown in FIG. **4**C is shown schematically as a Helmholtz resonator **190** in the center bullet **160**.

The treatments provided on the inner surface **147** may be any of these acoustic treatments, or any other acoustic treatments.

Possible wall treatments include passive types, such as fibrous, foam, ceramic, and filled cavity with a special handling of the treatment in a package.

Further, quarter-wavelength resonator type, Helmholtz resonators, single and double degree of freedom liners, made up of a perforated skin on honeycomb or partitions, volume tune liner types made up with fined tuned limited volumes on a top of which a perforated skin is bonded, where the perforations have a tuned size and number may all be used.

FIG. **5**A shows another exhaust duct embodiment **190** having an outer housing **246** with flow dividers **194**.

As shown in FIG. **5**B, there are crossing flow dividers **194** and **196** which are generally perpendicular to each other to define four separate chambers **292**. The flow dividers **194/196** and the inner wall **246** will each be provided with acoustic treatment as in the above embodiment.

FIG. **6** shows yet another embodiment **300** wherein there are plural flow dividers **350** which are generally parallel to each other and divide an interior of the outer exhaust wall **346** into three separate chambers **352**. Here again, acoustic treatments will be provided both on the outer surface of the flow dividers **350** and an inner surface of the outer wall **346** will provide effective noise reduction in the turbo-compounded engine.

The distance or the length of acoustically treated walls is a function of the level of attenuation needed. The total length is optimized along with the number of splitters and type of acoustic treatment.

A turbo-compounded engine **100** under this disclosure could be said to include a piston engine **116** connected to drive a propulsor **106**. An outlet **120** of the piston engine is operable to connect products of combustion from the piston engine **116** to pass over a turbine **122**. The turbine **122** is connected to drive a turbine shaft **121** also connected to drive the propulsor **106**. An outlet of the turbine is connected into an exhaust duct **146** configured to exhaust the products of combustion. The exhaust duct **146** is provided with an exhaust duct outer wall **146** defining an exhaust chamber **161**. A further cooling air outer wall **144** is positioned outwardly of the exhaust duct wall **146**. Flow dividers are received within an exhaust chamber inward of the exhaust duct outer wall. The exhaust duct outer wall has an inner surface and the flow dividers have an outer surface. Acoustic treatments are provided on both the inner surface of the exhaust duct outer wall and the outer surface of the flow dividers.

Although an embodiment has been disclosed, a worker of ordinary skill in this art would recognize that modifications would come within the scope of this disclosure. Thus, the following claims should be studied to determine the true scope and content of this disclosure.

What is claimed is:

1. A turbo-compounded engine comprising:

a piston engine connected to drive a propulsor, an outlet of the piston engine being operable to connect products of combustion from the piston engine to pass over a turbine, the turbine being connected to drive a turbine shaft also connected to drive the propulsor;

an outlet of the turbine connected into an exhaust duct configured to exhaust the products of combustion, the exhaust duct being provided with an exhaust duct outer wall defining an exhaust chamber, and a further cooling air outer wall positioned outwardly of the exhaust duct;

flow dividers received within the exhaust chamber inward of the exhaust duct outer wall, the exhaust duct outer wall having an inner surface and the flow dividers having an outer surface, and acoustic treatment being provided on both the inner surface of the exhaust duct outer wall and the outer surface of the flow dividers;

wherein the piston engine is a rotary engine;

wherein an electric motor is also provided to selectively drive the propulsor; and

wherein the exhaust duct outer wall has a first generally axially extending portion extending axially away from the engine with a component in an axial direction defined by the rotational axis of the engine, and the axial component being greater than a radial component, the first axial portion merging into a radially extending portion, and the radially extending portion merging into a second generally axially extending portion extending along a direction with a component in an axial direction that is greater than a radial component and the radially extending portion extending with an axial component but also having a radial component that is greater than the radial component in both the first and second generally axially extending portion.

2. A turbo-compounded engine comprising:

a piston engine connected to drive a propulsor, an outlet of the piston engine being operable to connect products of combustion from the piston engine to pass over a turbine, the turbine being connected to drive a turbine shaft also connected to drive the propulsor;

an outlet of the turbine connected into an exhaust duct configured to exhaust the products of combustion, the exhaust duct being provided with an exhaust duct outer wall defining an exhaust chamber, and a further cooling air outer wall positioned outwardly of the exhaust duct;

flow dividers received within the exhaust chamber inward of the exhaust duct outer wall, the exhaust duct outer wall having an inner surface and the flow dividers having an outer surface, and acoustic treatment being provided on both the inner surface of the exhaust duct outer wall and the outer surface of the flow dividers; and

wherein the exhaust duct outer wall has a first generally axially extending portion extending axially away from the engine with a component in an axial direction defined by the rotational axis of the engine, and the axial component being greater than a radial component, the first axial portion merging into a radially extending portion, and the radially extending portion merging into a second generally axially extending portion extending along a direction with a component in an axial direction that is greater than a radial component and the radially extending portion extending with an axial component but also having a radial component that is greater than the radial component in both the first and second generally axially extending portion.

3. The engine as set forth in claim **2**, wherein an electric motor is also provided to selectively drive the propulsor.

4. The engine as set forth in claim **3**, wherein a gear reduction is provided between an output shaft driven by the piston engine and the propulsor such that the propulsor rotates at a slower speed than the output shaft.

5. The engine as set forth in claim **4**, wherein a compound gearbox takes in rotation from the electric motor, the piston engine and the turbine and drives the output shaft leading into the gear reduction.

6. A turbo-compounded engine comprising:

a piston engine connected to drive a propulsor, an outlet of the piston engine being operable to connect products of combustion from the piston engine to pass over a turbine, the turbine being connected to drive a turbine shaft also connected to drive the propulsor;

an outlet of the turbine connected into an exhaust duct configured to exhaust the products of combustion, the exhaust duct being provided with an exhaust duct outer wall defining an exhaust chamber, and a further cooling air outer wall positioned outwardly of the exhaust duct;

flow dividers received within the exhaust chamber inward of the exhaust duct outer wall, the exhaust duct outer wall having an inner surface and the flow dividers having an outer surface, and acoustic treatment being provided on both the inner surface of the exhaust duct outer wall and the outer surface of the flow dividers;

the piston engine is a rotary engine;

wherein the exhaust duct outer wall has a first generally axially extending portion extending axially away from the engine with a component in an axial direction defined by the rotational axis of the engine, and the axial component being greater than a radial component, the first axial portion merging into a radially extending portion, and the radially extending portion merging into a second generally axially extending portion extending along a direction with a component in an axial direction that is greater than a radial component and the radially extending portion extending with an axial component but also having a radial component that is greater than the radial component in both the first and second generally axially extending portion;

wherein an electric motor is also provided to selectively drive the propulsor; and

wherein the flow dividers divide the exhaust chamber into at least three sub-chambers.

7. The engine as set forth in claim **6**, wherein the flow dividers include a central bullet and struts extending from the central bullet to the inner surface of the exhaust duct outer wall.

8. The engine as set forth in claim **6**, wherein the flow dividers are a plurality of plates extending to be perpendicular to each other to define the sub-chambers.

9. The engine as set forth in claim **6**, wherein the flow dividers are plural crossing plates which extend parallel to each other.

* * * * *